

Referee report

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Title: Z-Residuals for Diagnosing Bayesian Models

SUMMARY

The paper presents an approach to Bayesian model checking based on the notion of **Z-residuals**, which are defined for a given observation i as $z_i^{\text{RSP}}(y_i | \boldsymbol{\theta}) = -\Phi(\text{RSP}_i(y_i | \boldsymbol{\theta}))$, where $\Phi(\cdot)$ is the cdf of a standard normal and $\text{RSP}_i(y_i | \boldsymbol{\theta})$ are the *randomized survival probability*, defined as

$$\text{RSP}_i(y_i | \boldsymbol{\theta}) = S_i(y_i | \boldsymbol{\theta}) + U_i p_i(y_i | \boldsymbol{\theta}), \quad (1)$$

where $U_i \sim \text{Unif}(0, 1)$, $S_i(y_i | \boldsymbol{\theta})$ is the survival function of Y and $p_i(y_i | \boldsymbol{\theta})$ is its probability mass function (equal to zero if the random variable is continuous). Specifically, the authors show (in Theorem S1) that if the model is correctly specified, then $\text{RSP}_i(y_i | \boldsymbol{\theta}) \sim \text{Unif}(0, 1)$. In turn, this makes $z_i^{\text{RSP}}(y_i | \boldsymbol{\theta})$ normally distributed. In particular, the authors use this property to construct **posterior Z-residuals**, obtained by averaging $\text{RSP}_i(y_i | \boldsymbol{\theta}^{(t)})$ for each posterior sample $\boldsymbol{\theta}^{(t)}$. They also introduce a leave-one-out cross-validated version paired with importance sampling, to further correct the problematic double use of the data (which undermines calibration of the p-values). Their approach is useful for testing goodness-of-fit, e.g., via χ^2 test, as well as a tool to capture potential misspecifications in the functional form of a covariate. The authors showcase their approach in a variety of simulated settings and in a real dataset of dolphin-sighting.

COMMENTS

This is an interesting and potentially helpful contribution. While the notion of RSP is equivalent to the probability integral transform, which has been used for model checking in the Bayesian literature, what the author presents is a general strategy for checking both the correct specification of the likelihood and also which covariates might be misspecified. Moreover, the theoretical statements and proof appear correct. However, I have the following comments and questions.

1. The author presents differences between the RSP, MSP, and PPC. However, they should also compare their approach against the more recent holdout predictive checks (HPC) of [Moran et al. \(2023\)](#), which also avoid using the data twice and provide calibrated p-values, at the cost, however, of lower power. Since the proposed approach does not suffer from this loss of power, it would be great to see examples where your method works better (or comparatively).

2. It would be interesting to discuss your Z-residuals with the recent U-values posed by [Covington and Miller \(2025\)](#). Their approach introduces uniform distributions for each of the model parameters, while yours adds a z variable for each observation. Under which settings should your approach be preferred?
3. The authors indicate that the Z-residuals may detect potential misspecification in the functional form of some of the covariates (e.g. the simulation in Section 3.3). By looking at Figure 4.c, it looks like the trend between the z-residuals and x_1 in the model that assumes x_1 as linear has approximately a log-shape, which is the true functional form.
 - (a) The HNB family here is correctly specified. What would happen if one were to model the data as an NB model, under both with the correct functional form for x_1 (logarithmic) and the linear one? It would be important to clarify if the potential functional form between a given x and the z-residuals is impacted by the misspecification of the likelihood. A hint on what happens is suggested in Table 2 in the application (NB vs HNB in the nonlinear part), but since it is real data, it is hard to grasp the true relationship.
 - (b) Would it make sense to compare the z-residuals against a covariate that has *not* been used by the model?
4. The general style of the writing and the way the results are presented and discussed need improvement.
 - (a) I believe the authors should present the exact steps of their approach compactly in a simple algorithm. Since the authors are providing a general procedure, having a clear algorithm would significantly facilitate revisiting the paper, if one needs it.
 - (b) The paper contains a lot of abbreviations and combinations of hyphens (PPP, PPC, NB, HNB, Post-MSP, RSP_i^{CV} , $Z^{ISCV-RSP-SW}$ etc...), which are sometimes hard to follow. While these are summarized on the title page and have a rationale, the readability can be improved by limiting their use.
 - (c) The paper only provides a high-level summary of the theoretical results. It would be better to report the core of some of the statements (such as Theorem S2.2 and S2.5) in the main manuscript, while leaving the list of assumptions in the supplement.

REFERENCES

- Covington, C. T. and J. W. Miller (2025). Bayesian model criticism using uniform parametrization checks.
- Moran, G. E., D. M. Blei, and R. Ranganath (2023, 09). Holdout predictive checks for bayesian model criticism. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 86(1), 194–214.